October University for Modern Sciences and Arts (MSA)
Department of Electronics and Communication
Course: ECE552 Mobile Communication Systems
Instructors: Prof. Fawzy Ibrahim Dr. Abdelrahman El-Akhdar
Assoc. Prof. Nariman

Problem Set #3 Solution

Question #3.1

Free space propagation: Assume the transmitter power is 1 w at 60 GHz fed into the transmitter antenna. Using the horn antenna; with dimensions 4.6 cm * 3.5 cm; at both the transmitter and receiver:

a) Calculate the free space path loss at 1 m, 100 m, and 1000 m.
b) Calculate the received signal power at these distances.

Solution:

a) For free space propagation loss:

FSPL = 32.4 + 20 log d (km) + 20 log f (MHz)

FSP (d = 1 m) $= 32.4 + 20 \log (10^{-3}) + 20 \log (60*10^{3}) = 67.96$ dB.
FSPL (d = 100 m) $= 32.4 + 20 \log (100*10^{-3}) + 20 \log (60*10^{3}) = 107.96$ dB.
FSPL (d = 1000 m) $= 32.4 + 20 \log (1) + 20 \log (60*10^{3}) = 127.96$ dB.

P_r (dBw) = P_t (dBw) + G_t (dBi) + G_r (dBi) – FSPL (dB)

$G_t = G_r = (4*pi*Ae) / \lambda^2$; **$A_e = e_o$ Ag** and **e_o** = 60 – 75 %
For horn antenna **e_o** = 1
$\lambda = c / f = 5*10^{-3}$ m
$A_e = (4.6*10^{-2} * 3.5*10^{-2}) = 1.61*10^{-3}$ m^2.
$G_t = G_r = 4\pi A_e / \lambda^2 = 809.27$
G_t (dBi) = G_r (dBi) = 10 log (809.27) = 29.08 dBi

$P_t = 10 \log (1) = 0$ dBw

d = 1 m: P_r (dBw) = P_t (dBw) + G_t (dBi) +G_r (dBi) – FSPL (dB) = -9.8 dBw

d = 100 m: P_r (dBw) = P_t (dBw) + G_t (dBi) +G_r (dBi) – FSPL (dB) = -49.8 dBw

d = 1000 m: P_r (dBw) = P_t (dBw) + G_t (dBi) +G_r (dBi) – FSPL (dB) = -69.8 dBw

Question #3.2 **[HW]**

Free space propagation: Assume the transmitter power is 2 w at 10 GHz fed into the transmitter antenna. Using the horn antenna; with dimensions 6 cm * 5 cm; at both the transmitter and receiver:

a) Calculate the free space path loss at 10 m, 100 m, and 1000 m.
b) Calculate the received signal power at these distances.

Solution as in Question #3.1

Question #3.3

GSM mobile communication network is designed to work in the downtown area with radius, R_{GSM} = 2 km of coverage for each site and the working frequency f_{GSM} = 940 MHz. The system is upgraded to work as 3G (UMTS) with identical system operational parameters but with double coverage distance, $R_{UMTS} = 2\ R_{GSM}$ = 4 km and the working frequency f_{UMTS} = 2100 MHz. If the Base Station (BS) antenna height is h_{ta} = 35 m and the Mobile Station (MS) antenna height is h_{ra} =1 m Find:

a) Power reduction factor by assuming free space path loss model.
b) Power reduction factor by assuming practical path loss models.

Comment on your results

Area	**Okumura-Hata Model**	**Hata - COST Model**
Urban	$L_{50} = 69.55 + 26.16 \log (f_c) - 13.82\log (h_{te}) – a(h_{re}) + [44.9 - 6.55 \log (h_{te}) \log (d)]$	$L_{50} = 46.3 + 33.9 \log (f_c) - 13.82 \log (h_{te}) - a(h_{re}) + [44.9 - 6.55\log (h_{te})] \log (d) + C_M$
Small, downtown, and Medium City	$a(h_{re}) = (1.11 \log (f_c) - 0.7)\ h_{re} - (1.56 \log (f_c) - 0.8)$ dB	
Large City	$a(h_{re}) = 8.29\ [\log(1.54h_{re})]^2 - 1.1$dB for $f_c <= 1500$MHz $a(h_{re}) = 3.2\ [(\log(11.75h_{re})]^2 - 4.97$dB for $f_c => 1500$MHz	
Sub-Urban	$L_{50}(dB) = L_{50}(Urban) - 2[\log(f_c/28)]^2 - 5.4$	
Open and rural area	$L_{50}(dB) = L_{50}(Urban) - 4.78\ [\log (f_c)]^2 + 18.33 \log (f_c) - 40.94$	
C_M for Hata – COST Model	$C_M = 0$ dB for Medium and Sub-Urban, $C_M = 3$ dB for metropolitan centers and downtown	

Solution:

Double coverage distance (R_{GSM} = 0.5 and R_{UMTS} = 2Km)

a) Free space Path loss model:

For GSM:

$P_r = P_t(GSM) + G_t + G_r - FSPL(GSM)$ and

$FSPL(GSM) = 32.4 + 20 \log (f_{GSM}) + 20 \log(R_{GSM})$

$P_r(GSM) = P_t(GSM) + G_{ta} + G_{ra} - [32.4 + 20 \log (f_{GSM}) + 20 \log(R_{GSM})]$

For UMTS:

$P_r(UMTS) = P_t(UMTS) + G_{ta} + G_{ra} - FSPL(UMTS)$ and

$FSPL(UMTS) = 32.4 + 20 \log (f_{UMTS}) + 20 \log (R_{UMTS})$

$P_r = P_t(UMTS) + G_{ta} + G_{ra} - [32.4 + 20\log (f_{UMTS}) + 20\log (R_{UMTS})]$

So for the same P_r, assume identical operational parameters, so we get:

$P_t(GSM) + G_{ta} + G_{ra} - [32.4 + 20 \log (f_{GSM}) + 20 \log(R_{GSM})] =$

$P_t(UMTS) + G_{ta} + G_{ra} - [32.4 + 20\log(f_{UMTS}) + 20\log (R_{UMTS})]$

Power reduction factor $= P_t(UMTS) - P_t(GSM)$

$= 20 \log (f_{UMTS}/f_{GSM}) + 20 \log (2) =$

$= 20 \log (2100/940) + 20 \log (2) = 13$ dB

b) Practical Path Loss Models:

For GSM:

$P_r(GSM) = P_t(GSM) + G_{ta} + G_{ra} - L_{50GSM}$

and L_{50GSM} is the Hata – COST Model (f_{GSM} is less than 1500MHz)

$L_{50GSM} = 137.15$dB

$P_r(GSM) = P_t(GSM) + G_{ta} + G_{ra} - 137.15$

For UMTS:

$P_r(UMTS) = P_t(UMTS) + G_{ta} + Gra - L_{50UMTS}$

and L_{50UMTS} is the Hata – COST (f_{UMTS} is greater than 1500 MHz)

$L_{50UMTS} = 162.15$ dB

$P_r(UMTS) = P_t(UMTS) + G_{ta} + G_{ra} - 162.15$

So for the same P_r, assume identical operational parameters, so we gat:

$P_t(GSM) + G_{ta} + G_{ra} - 137.15 = P_t(UMTS) + G_{ta} + G_{ra} - 162.15$

Power reduction factor = $P_t(UMTS) - P_t(GSM) = 25$ dB

Comment on the obtained results

We can predict that the practical models is much accurate than the free space model because of including the different channel parameters.

Question #3.4 **[HW]**

GSM (DCS)1800 mobile communication network is designed to work in the downtown area with radius, R_{DCS} = 2.5 km of coverage for each site and the working frequency, f_{DCS} = 1800 MHz. The system is upgraded to work as 3G (UMTS) with identical system operational parameters but working frequency f_{UMTS} = 2100 MHz. If the Base Station (BS) antenna height is h_{ta} = 40 m and the Mobile Station (MS) antenna height is h_{ra} =1.5 m Find:

a) The new Cell radius, R_{UMTS} by assuming free space path loss model.

b The new Cell radius, R_{UMTS} by assuming Hata - COST propagation model.

Solution: The same steps as Question #3.3.

Question #3.5

Four received power measurements were taken at distances d_1 = 100 m, d_2 = 200 m, d_3 = 1 km, and d_4 = 4 km from a transmitter. The measured values at these distances are: P_{m1} = 0 dBm, P_{m2} = -24 dBm, P_{m3} = -30 dBm, and P_{m4} = -59 dBm, respectively. It is assumed that the path loss for these measurements follows the log-normal model where, d_o = 100 m.

a) Find the Mean Square Error (MSE) estimate for the path loss exponent n.

b) Estimate the received power at d = 3.5 km using the resulting model.

Solution:

a) Squared Error is defined by: $E(n) = \sum_{i=1}^{K} (P_{rn} - P_{re})^2$

But $P_{ri} = P(d_o) - 10n \log(\frac{d_i}{d_o})$, assume $P(d_0)=0$

So, $P_{r1} = -10n \log(\frac{d_1}{d_o})$, $P_{r2} = -10n \log(\frac{d_2}{d_o})$, …, $P_{rK} = -10n \log(\frac{d_K}{d_o})$

Hence

$E(n) = (P_{m1} + 10n \log(\frac{d_1}{d_o}))^2 + (P_{m2} + 10n \log(\frac{d_2}{d_o}))^2 + \ldots\ldots + (P_{mn} + 10n \log(\frac{d_n}{d_o}))^2$.

For MSE we have; $\frac{dE(n)}{dn} = 0$

$2(P_{rm1} + 10n \log(\frac{d_1}{d_o}))(10\log(\frac{d_1}{d_o})) + 2(P_{rm2} + 10n \log(\frac{d_2}{d_o}))(10\log(\frac{d_2}{d_o})) + \ldots. +$

$2(P_{rmn} + 10n\log(\frac{d_n}{d_o}))(10n \log(\frac{d_n}{d_o})) = 0$

b) For the case of having only 4 measured values: $n = -\frac{\sum_{i=1}^{4} P_{rmi} \log \frac{d_i}{d_\circ}}{10\sum_{i=1}^{4}\left(\log \frac{d_i}{d_\circ}\right)^2}$ So, n= 3.6 ;

$$P_r(3.5\text{Km}) = -10n\log\left(\frac{d_x}{d_o}\right) = -55.6\text{dBm} = 2.76\ \text{nW} = 0.00276\ \mu\text{W}$$

Question #3.6 **[HW]**

Four received power measurements were taken at distances d_1 = 100 m, d_2 = 500 m, d_3 = 1 km, and d_4 = 2 km from a transmitter. The measured values at these distances are: P_{m1} = 0 dBm, P_{m2} = -25 dBm, P_{m3} = -30 dBm, and P_{m4} = -36 dBm, respectively. It is assumed that the path loss for these measurements follows the log-normal model where, d_o = 100 m.

a) Calculate the Mean Square Error (MSE) estimate for the path loss exponent n.

b) Estimate the received power at d = 5 km using the resulting model.